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Characterization results of the JUNGFRAU full scale readout ASIC

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ABSTRACT: The two-dimensional pixel detector JUNGFRAU is designed for high performance photon science applications at free electron lasers and synchrotron light sources. It is developed for the SwissFEL currently under construction at the Paul Scherrer Institut, Switzerland. The detector is a hybrid pixel detector with a charge integration readout ASIC characterized by single photon sensitivity and a low noise performance over a dynamic range of 10^4 12 keV photons.

Geometrically, a JUNGFRAU readout chip consists of 256×256 pixels of $75 \times 75 \mu\text{m}^2$. The chips are bump bonded to $320 \mu\text{m}$ thick silicon sensors. Arrays of 2×4 chips are tiled to form modules of $4 \times 8 \text{ cm}^2$ area. Several multi-module systems with up to 16 Mpixels per system will be delivered to the two end stations at SwissFEL.

The JUNGFRAU full scale readout ASIC and module design are presented along with characterization results of the first systems. Experiments from fluorescence X-ray, visible light illumination, and synchrotron irradiation are shown. The results include an electronic noise of ~ 50 electrons r.m.s., which enables single photon detection energies below 2 keV and a noise well below the Poisson statistical limit over the entire dynamic range. First imaging experiments are also shown.

KEYWORDS: X-ray detectors; Pixelated detectors and associated VLSI electronics; Solid state detectors; Hybrid detectors

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Introduction

A new generation X-ray source, the Swiss Free Electron Laser (SwissFEL), is currently being built at the Paul Scherrer Institut (PSI, Villigen, Switzerland). The facility will be operational in 2017 and will generate femtosecond pulses with up to 0.7×10^{11} photons per pulse with a photon energy in the 2-12 keV range [1] in its hard X-ray ARAMIS beamline.

In parallel to the development of the light source, PSI is developing a two-dimensional pixel detector, called JUNGFRAU (adJUsTiNg Gain detector FoR the Aramis User station), that is specifically tailored to the needs of the machine itself.

The detector is based on the hybrid pixel detector technology, where a high resistivity silicon sensor is bonded to a custom designed readout application-specific integrating circuit (ASIC). In case of the JUNGFRAU project, the ASIC is characterized by single photon sensitivity and a low noise performance over a dynamic range of 10^4 12 keV photons. These distinguishing characteristics are achieved by a three-fold, gain-switching preamplifier in each pixel, which dynamically adjusts its gain to the amount of charge deposited on the pixel, an architecture developed at PSI for the AGIPD [2, 3] and GOTTHARD [4] projects and used in some variations of the ePix detector class [5].

The final, full size, readout ASIC has been recently fabricated and bump bonded to sensors, both in single chip and in 8-chip module configurations. The characterization of the ASIC and the module performance is the topic of this work.

Sections 1, 2 and 3 introduce the design of the ASIC, of the module and of the readout system respectively. Section 4 reports the results of the noise characterization in high gain or single photon regime. Section 5 and section 6 show the results of the dynamic range and uniformity tests.

1 The JUNGFRAU 1.0 readout ASIC

The JUNGFRAU 1.0 full scale readout ASIC is designed in UMC¹ 110 nm CMOS technology with Aluminum-only interconnect. The pixel architecture of the JUNGFRAU 0.1 and 0.2 prototype systems, which has been described in detail in [6] and in [7], has been used with small modifications in this full scale design and will not be further discussed in this paper. In summary, it consists of the the following blocks: *a preamplifier* with 3 gains, *an automatic gain switching block* consisting of a comparator with tunable threshold and a switching control logic, *a correlated double sampling* (CDS) stage to remove the preamplifier low frequency and reset noise in high gain, *a storage array* for up to 16 images, *a voltage buffer* which drives the column bus during the readout phase.

In addition to the three gains of the standard switching sequence (denominated, from high to low, G0, G1, and G2), an additional, globally selectable operation mode can increase the gain of high gain setting to High G0 or HG0. This reduces the noise and allows single photon detection at lower energies; as a trade-off the gain ratios in the new switching sequence (HG0, G1, G2) are not optimal, the first gain having a small range.

The pixels are arranged on a 75 μm pitch 256 $\times$ 256 array, which sits very close ($<20\ \mu\text{m}$) to the chip edge on the left, top and right sides. Apart from the row shift register on the left side of the chip, the periphery is concentrated at the bottom of the pixel array.

The periphery has two main functions: it provides buffering for the digital signals and biases which are fed into the pixel array and it performs the multiplexed analog and digital readout of the pixel outputs. The latter task is organized as follows: every pixel column has, at its bottom, a voltage column amplifier which buffers the pixel output, operated at low speeds, and provides low impedance driver for the following high speed multiplexer and output buffer stages. The pixel matrix is organized in 4 super-columns, 64 pixels wide, each with its own multiplexer and fully differential off-chip driver. This part of the readout chain has been designed to allow a direct drive of the external Analog to Digital Converters (ADC) at a readout frequency up to 40 MHz, corresponding to a global frame rate of 2.4 kHz.

The nominal power consumption of the ASIC is about 2 W, with most of the power used by the pixel array. Each pixel drains 10 μW (at 1.2V) for the preamplifier and 15–20 μW (at 1.4 V) for the correlated double sample stage and comparator. The latter power consumption can be defined and tuned by means of external Digital to Analog Converter channels. The preamplifier current is fixed, and was measured in silicon to be within 15% of the design specifications.

2 Multi ASIC module design

Eight JUNGFRAU 1.0 readout chips are bump bonded in a 4 $\times$ 2 configuration to a single 320 μm thick silicon sensor from Hamamatsu.² In the regions of the gaps between two ASICs 2 rows or columns with double-sized pixels are present on the sensor to cover the gaps themselves. Where the corners of the ASICs meet, four pixels four times the area are present on the sensor. In total, 1024 $\times$ 512 pixels are present, covering an area of 77.25 $\times$ 38.55 mm².

¹United Microelectronics Corporation.

²<http://www.hamamatsu.com/>.

After bump bonding, the ASIC+sensor assembly is glued to the High Density Interconnect (HDI) PCB, which is itself glued to an Aluminum module mechanical block providing geometric stability and thermal connection to the detector cooling plate. The HDI handles the electrical interconnection between the ASIC pads and a 140 pin connector, where the readout electronic board is inserted.

In the final step the ASICs are wire bonded, with a $25\mu\text{m}$ Aluminum wire, to the HDI board; high voltage connection to the sensor backplane is provided as well through a wire bond to the HDI.

The modules can be tiled in arbitrary arrays. As a result of a careful optimization of the mechanical assembly, inherited from the EIGER project [8], the gaps between modules are very small: 2.7 mm on the long side and 0.6 mm on the short one, pixel to pixel. This corresponds to 36 and 8 missing pixels respectively, which amounts to a total active area of 93% on a unit cell (97% on a 2M system).

The results shown in this paper have been obtained with the first operational JUNGFRU assembly ever produced. No wafer level ASIC testing was performed at this stage, so not all ASICs on the module were guaranteed to work. For this reason, a low quality sensor, i.e. a sensor with an out-of-specification number of visible defects, has also been used.

3 The readout system

The results presented in the following sections have been obtained with a prototype version of the JUNGFRU Module Control Board (JMCB), which is the main component of a JUNGFRU module. The JMCB main component is an Altera CycloneV Field Programmable Gate Array (FPGA) with transceivers. The FPGA generates the digital signals to control the ASICs; during the ASIC readout phase the digital signals (gain bits) are directly latched in the FPGA itself, while the multiplexed analog output from the pixels is first digitized by a set of AD9257 14 bit ADCs. A total of 32 ADC channels operating at a maximum frequency of 40 MHz are present on the board. The stream of data (digital+analog) is properly descrambled to obtain a contiguous image which is temporary stored on an on-board DDR3 memory. The images in memory can be streamed out to the user server via up to two 10 Gbit fiber links with a UDP protocol. With both links active, the system is designed to sustain readout frame rates higher than 2 kHz.

The configuration and slow control of the system are handled by a system-on-a-board embedded Linux system³ with a Blackfin CPU. The embedded processor is controlled from the user client PC via a custom TCP-IP socket interface over 100 MBit/s Ethernet.

The power to the JMCB is provided by a single 12 V connector, and DC-DC converter plus linear regulator chains are used to generate board and ASIC voltages. The high voltage sensor bias (up to 200 V) is generated by a small form factor daughter board with a digital control of the output voltage.

The total power consumption of the module during measurements was ~ 30 W, 2/3 of which are used by the ASICs themselves; the cooling is provided by a water loop, designed to have a high thermal conductivity to the ASIC+sensor assembly.

Unless differently specified, the following system parameters have been used during this characterization work:

³<http://www.bluetechnix.com>.

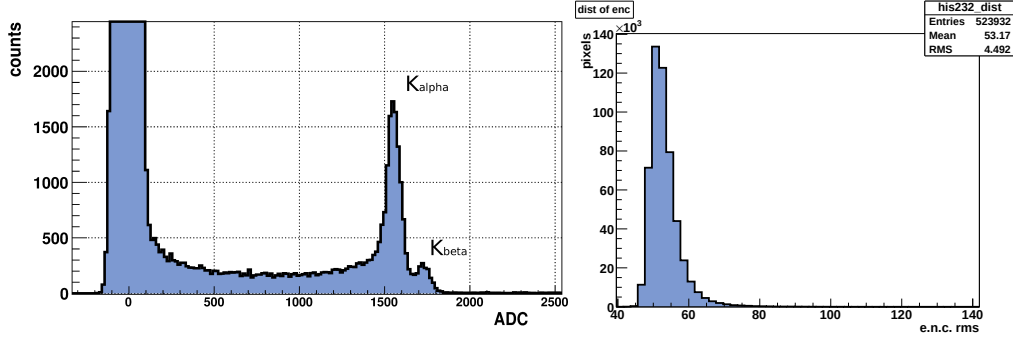


Figure 1. Left: pulse Height distribution of a sample pixel for a Cu fluorescence irradiation in HG0. The K_α and K_β lines of Cu, at 8.05 keV and 8.90 keV respectively, are clearly resolved. The pixel has been chosen from the MPV pixels of the noise distribution. Right: noise distribution in equivalent noise charge for the 525k pixels of a module.

- streaming frames at 700 Hz with one 10 Gbit link, limited by the prototype firmware;
- 20 MHz ASIC and ADC readout, corresponding to 1.2 kHz ASIC frame rate;
- 20°C water cooling temperature, ~40°C ASIC temperature;
- 200 V sensor high voltage, onboard generated.

4 Noise at high gain

To measure the noise at high gain, i.e. in single photon regime, the detector has been irradiated with X-ray fluorescence light (from different elemental samples) generated by an X-ray tube at PSI.

An integration time of 10 μ s, which is close to the 2-5 μ s of the expected XFEL operation but provides a higher occupancy and thus a faster data collection, was selected.

For each measurement a large number of frames has been collected, with and without photons: the ones without photons have been used to compute the pixel-by-pixel offset which takes into account variations in the electronic devices and the sensor leakage current. This “dark image” is then subtracted from the photon data. The resulting pulse height distribution for a sample pixel and Cu fluorescent light irradiation, is shown in figure 1(left). In the plot are visible, from left to right: the noise peak at 0 ADC, a plateau due to charge sharing between two neighbor pixels, and the K_α , K_β fluorescence lines of Copper. A fit on the Gaussian photon peak of such pulse height distribution allows to calibrate the pixel gain of the system in terms of eV/ADC; the equivalent noise charge (e.n.c.) can be computed simply from the r.m.s. of the output distributions in absence of any photon signal.

The gain and noise profiles can be extracted by fitting the pulse height distributions of all pixels: the resulting distribution of the noises is shown, for HG0 in figure 1(right). The Most Probable Value (MPV) of the e.n.c. distribution is 50 e_{rms}^- , and the average is 53 e_{rms}^- , with only a few pixels with a noise greater than 80 e_{rms}^- . The result exceeds the design specifications and will allow measurements with single photon resolution below a 2 keV photon energy.

The low energy single photon resolution capabilities have been verified during a dedicated experiment performed at the PHOENIX beamline, Swiss Light Source, PSI. A single ASIC

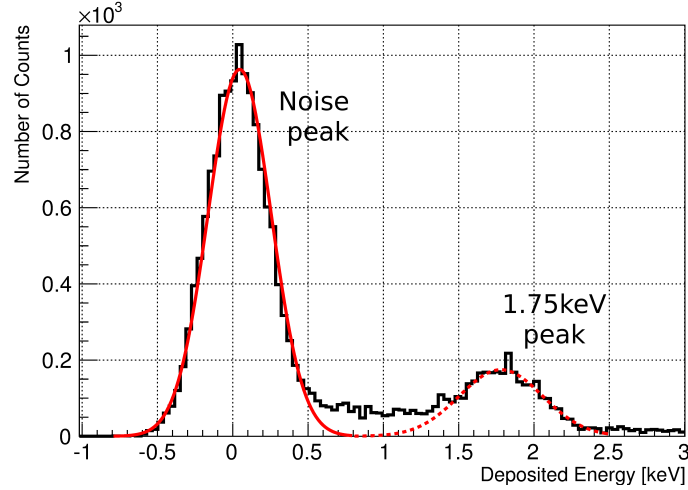


Figure 2. Pulse Height distribution, in HG0, of a sample pixel under a 1.75keV photon irradiation at the SLS PHOENIX beamline. The energy scan data was used to convert the raw ADC unit output into keV. A fit with two Gaussians is superimposed.

JUNGFRAU assembly has been placed inside the PHOENIX vacuum chamber. Extension cables with vacuum ports have been used between the assembly and the readout board. No measurable degradation of performance, e.g. increase in noise, is introduced by the 50 cm long cables. The high voltage has been increased to 350V to enhance the charge collection properties in the region, close to the backplane, where the low energy photons mainly convert. The system has been irradiated with an attenuated, non-focused direct beam, while the energy has been varied in the 1.75-3 keV range. The result, for a sample pixel and a 1.75 keV beam energy is presented in figure 2. The photon peak is very well resolved and the offline photon discrimination can still use a half energy threshold, which gives the best results in terms of spatial uniformity. The noise in G0 has been measured with a very similar procedure, and is $83 \pm 6 e_{\text{rms}}^-$. This gain mode can thus safely be used for photon energies down to 3 keV. Gain uniformities within 3% r.m.s. have been measured for both G0 and HG0 modes. The gain fluctuations are mainly due to the power distribution and to the pixel readout sequence, since they do not appear to be randomly distributed but can be associated to gain gradients along the chip. During beamline operation, a pixel by pixel gain map will have to be applied to the raw data.

5 Dynamic range scan

A visible light source was used to test the noise and range figures of merit in gain switching operation, following [9]. For this purpose a hole was etched in the aluminum backplane metalization of the silicon sensor to allow visible light to reach the sensitive region. The input charge was modulated by controlling the integration time. For each integration point, several images were collected and the mean and r.m.s. of the pixel output voltage were computed. The results, after conversion of the input charge into number of 12 keV photons with the help of the previously measured gain factors, are shown in figure 3 (Top). The change in slope sign between the 1st and the 2nd gain is due to the CDS stage removal after gain switching. In figure 3 (Bottom) the r.m.s. noise values converted into 12 keV photon units are represented; the noise values are always much smaller than the Poisson

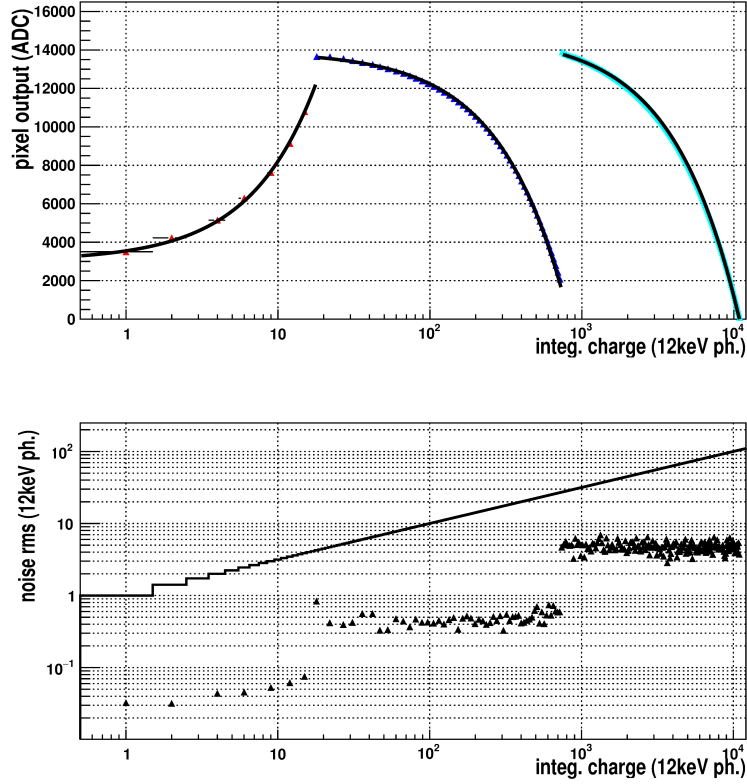


Figure 3. Output response (top) and r.m.s. noise values (bottom) of a sample pixel as a function of the incoming integrated charge, on a logarithmic scale. Linear fits are superimposed to each gain response. A line representing the fluctuations due to Poisson statistics is added to the noise plot.

Table 1. Dynamic range and noise for the different gains in automatic switching operation. The cases of 12 keV and 2 keV photon energies, where G0 and HG0 modes would be selected respectively, are reported.

Gain	G0 (12 keV photon energy)			HG0 (2 keV photon energy)		
	range	noise	Poisson fluct.	range	noise	Poisson fluct.
G0/HG0	0-20 γ	83 e^-_{rms}	-	0-60 γ	53 e^-_{rms}	-
G1	20-700 γ	0.36 γ	> 4.5 γ	60-4200 γ	2 γ	> 7.7 γ
G2	700-11k γ	4 γ	> 26 γ	4200-66k γ	25 γ	> 65 γ

statistical fluctuations on the number of photons, as specified in the design requirements. The noise is therefore limited by statistical fluctuations in the number of photons. Table 1 shows the noise and range parameters of the three gain stages for two photon energy cases of 12 keV and 2 keV, which correspond to the high and low ends of the photon energy range of the SwissFEL hard X-ray beamline respectively. The Poisson fluctuation on the number of photons refers to the critical point, i.e. after each gain switching, where electronic noise is closer to the statistical fluctuation.

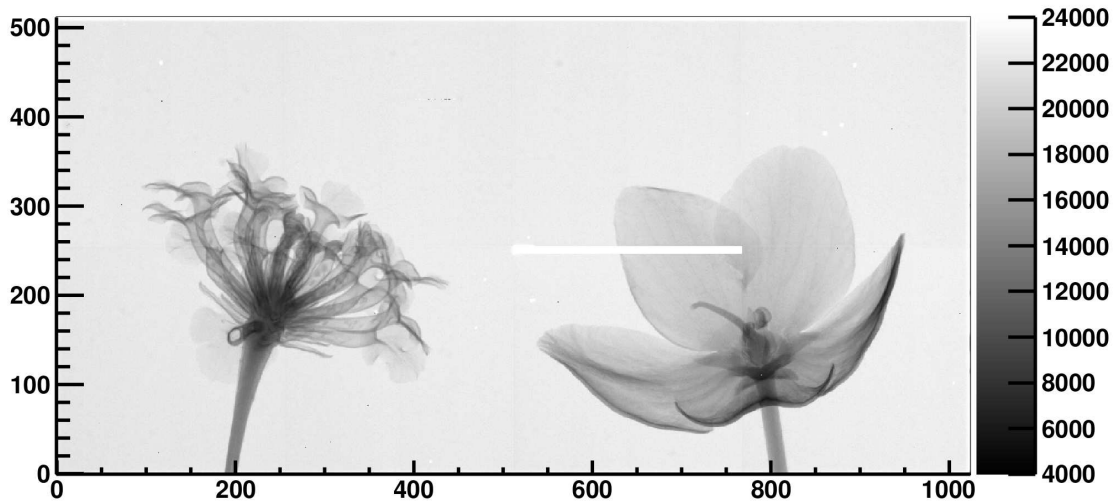


Figure 4. X-ray image of two flowers, with an a X-ray tube setup. See text for details on experimental conditions and data analysis.

6 Example of imaging

Some X-ray radiographies have been collected with the JUNGFRU module. Although this kind of images do not provide much quantitative information on the system performances, they provide a qualitative assessment on the global uniformity and yield of the system.

The object, two delicate flowers, has been placed in front of the direct beam from a Ag anode, with a 1 mm spot size, operated at 45 kV, 1 mA. The ratio between source-sample and sample-detector distances was ~ 50 . The detector has been operated in single photon regime, i.e. one or fewer photon per pixel per frame, at a 700 Hz frame rate and 10 μ s integration time. After a pedestal correction, an offline photon counter algorithm has been applied to the 100k frames data-set, with an ADC unit threshold corresponding to 5 keV. No gain or flat field corrections have been applied.

The result is shown in figure 4: one row (hardly visible) in chip #2 and 10 rows in chip #7⁴ are completely not responding. These defects are expected since no wafer level chip testing was performed. This considered, the module shows a good uniformity. A few dead regions 3 to 8 pixel wide (white spots) are present; these are related to defects in the Indium ball deposition on the out-of-specification sensor which has been used. The overall yield is nevertheless better than 99.5%, or 99.95% if the bad rows due to the (untested) ASIC are accounted for.

7 Conclusions

The design and fabrication of the full scale JUNGFRU 1.0 readout ASIC has been completed, and the first modules have been produced. The systems have been characterized with X-ray fluorescence beams, low energy direct beams at the SLS PHOENIX beamline, and visible light scans. The results of the tests are promising. In particular, an electronic noise as low as 50 e^-_{rms} has been measured in HG0 mode, with standard settings and with a full 500k pixel module. This figure of merit is a factor

⁴Chips are numbered left to right and top to bottom, as a written text.

of 2 lower than in the JUNGFRAU 0.2 prototype [7] and will enable single photon detection for energies below 1.75 keV. Dynamic range, noise across the range, and uniformity have been measured to be within design specification. The results in terms of module quality and bump bond yield are encouraging and indicate that a mass production run of modules in a limited time frame will be possible.

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References

- [1] R. Ganter et al., *SwissFEL conceptual design report*, [PSI Bericht 10-04](#), Paul Scherrer Institut, Villigen Switzerland (2010).
- [2] X. Shi et al., *Challenges in chip design for the AGIPD detector*, [Nucl. Instrum. Meth. A](#) **624** (2010) 387.
- [3] B. Henrich et al., *The adaptive gain integrating pixel detector for the European XFEL*, [Nucl. Instrum. Meth. A](#) **633** (2011) S11.
- [4] A. Mozzanica et al., *A single photon resolution integrating chip for microstrip detectors*, [Nucl. Instrum. Meth. A](#) **633** (2011) S29.
- [5] A. Dragone et al., *ePix: a class of front-end ASICs for second generation LCLS integrating hybrid pixel detectors*, [IEEE Nucl. Sci. Symp. Conf. Rec.](#) (2013) 1.
- [6] A. Mozzanica et al., *Prototype characterization of the JUNGFRAU pixel detector for SwissFEL*, [2014 JINST](#) **9** C05010.
- [7] J. Jungmann-Smith et al., *JUNGFRAU 0.2: prototype characterization of a gain-switching, high dynamic range imaging system for photon science at SwissFEL and synchrotrons*, [2014 JINST](#) **9** P12013.
- [8] R. Dinapoli et al., *EIGER: next generation single photon counting detector for X-ray applications*, [Nucl. Instrum. Meth. A](#) **650** (2011) 79.
- [9] A. Mozzanica et al., *The GOTTHARD charge integrating readout detector: design and characterization*, [2012 JINST](#) **7** C01019.